

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

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Task 1: PEAS Specification

Selected Application: An Agricultural Monitoring Drone for crop disease detection.

Performance Measure:

1. Disease detection accuracy: Percentage of correctly identified diseased plants versus healthy plants.
2. False positive rate: Some healthy plants misidentified as diseased, avoiding unnecessary pesticide use.
3. Coverage efficiency: Acres of farmland successfully scanned per hour.
4. Battery utilization: Efficient flight time and area covered per full charge cycle.

Environment:

The drone operates in large open agricultural farmlands and crop fields. Key external factors include unpredictable weather conditions (wind, rain, sunlight glare), varying crop heights, and the presence of physical obstacles like trees, birds, or power lines.

Actuators:

1. Quadcopter flight propellers and motors for movement and stabilization.
2. Adjustable camera gimbal to direct sensors at specific crops.
3. Optional localized pesticide sprayer for immediate treatment of identified diseases.

Sensors:

1. High-resolution RGB cameras for visual crop inspection.
2. Multispectral and Infrared cameras to detect hidden plant stress or disease.
3. GPS module for precise geolocation and mapping.
4. LiDAR and Ultrasonic sensors for altitude control and obstacle avoidance.

Implementation Note: For the purpose of system development and lab evaluation, all physical sensors and actuators mentioned above can be substituted with a software simulation environment.

Task 2: Environment Classification

Fully Observable vs. Partially Observable: Partially Observable.

Justification: Occlusion from plant leaves, overlapping canopy, and changing weather conditions hide some crop regions and prevent full visibility of the entire field at once.

Single-agent vs. Multi-agent: Single-agent.

Justification: The drone operates independently as the primary decision-maker to monitor its designated field sector without needing to negotiate or compete with other AI agents.

Deterministic vs. Stochastic: Stochastic.

Justification: External natural factors like sudden wind gusts, lighting changes, and slight battery fluctuations make the exact outcome of physical flight actions somewhat unpredictable.

Episodic vs. Sequential: Sequential.

Justification: The drone's current path planning and battery usage directly constrain and impact the areas it will be able to scan in future steps.

Static vs. Dynamic: Dynamic.

Justification: Weather conditions, sunlight intensity, and moving obstacles like birds or farm equipment change constantly while the drone is in flight deciding its next move.

Discrete vs. Continuous: Continuous.

Justification: The drone's physical state (its position coordinates, altitude, speed, and camera angles) operates smoothly within a continuous physical space and time.

Task 3: Critical Analysis

1. Greatest Technical Challenge:

The greatest technical challenge for an AI engineer building this system is the Stochastic and Dynamic nature of the environment. Unlike controlled indoor settings, an open farmland involves unpredictable, rapid changes in weather (e.g., wind gusts pushing the drone, clouds causing extreme lighting variations) and moving obstacles. Designing a computer vision system that remains highly accurate despite these lighting changes, combined with a flight controller that can instantly stabilize the drone against unpredictable wind, requires highly robust, real-time sensor fusion algorithms.

2. Utility Function and Trade-offs:

Utility Function: $U = (w_1 * \text{Area_Covered_Per_Hour}) - (w_2 * \text{Collision_Risk})$

This utility function helps the agent balance the trade-off between Speed (Coverage) and Safety. If the weight of the safety parameter (w_2) was doubled, the drone's behavior would become highly conservative. It would fly at lower speeds, maintain a higher altitude, and keep a wider berth from any detected obstacles like trees or power lines. While this would decrease its overall Area_Covered_Per_Hour, it would significantly minimize the likelihood of a crash, ensuring the expensive equipment is kept safe in unpredictable weather.

Structured Representation:

```
{
  "system_name": "Agricultural Monitoring Drone",
  "peas": {
    "performance_measures": [
      "disease_detection_accuracy",
      "false_positive_rate",
      "coverage_efficiency_acres_per_hour",
```

```

    "battery_utilization_flight_time"
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  "environment": "Large open farmlands with unpredictable weather, crops, and obstacles",
  "actuators": [
    "quadcopter_propellers",
    "adjustable_camera_gimbal",
    "localized_pesticide_sprayer"
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  "sensors": [
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    "GPS_module",
    "LiDAR_sensor"
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    "choice": "Partially Observable",
    "justification": "Occlusion from plant leaves and changing weather hide crop regions."
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    "choice": "Single-agent",
    "justification": "The drone operates independently without negotiating with other AI agents."
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  "determinism": {
    "choice": "Stochastic",
    "justification": "Wind gusts and lighting changes make flight outcomes somewhat unpredictable."
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  "episodicity": {
    "choice": "Sequential",
    "justification": "Current path planning directly impacts future scanning capabilities."
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  "dynamism": {
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    "justification": "Weather, lighting, and moving obstacles change while making decisions."
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  "continuity": {
    "choice": "Continuous",
    "justification": "Drone position and altitude operate in continuous physical space."
  }
},
"utility_function": "U = (w1 * Area_Covered_Per_Hour) - (w2 * Collision_Risk)"
}

```